

# Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Data Cohort Shift

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## Highlights

- Positive loss weighting induces an additive population logit offset of  $\log w$ .

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- Finite neural network logits empirically cluster tightly around theoretical  $\log w$ .
- Temperature scaling cannot correct this intercept-type calibration bias.
- Analytic offset ( $z - \log w$ ) cuts 73.7% of Brier error without retraining.
- Selective triage exposes trade-offs between calibrated risk and screening referral.